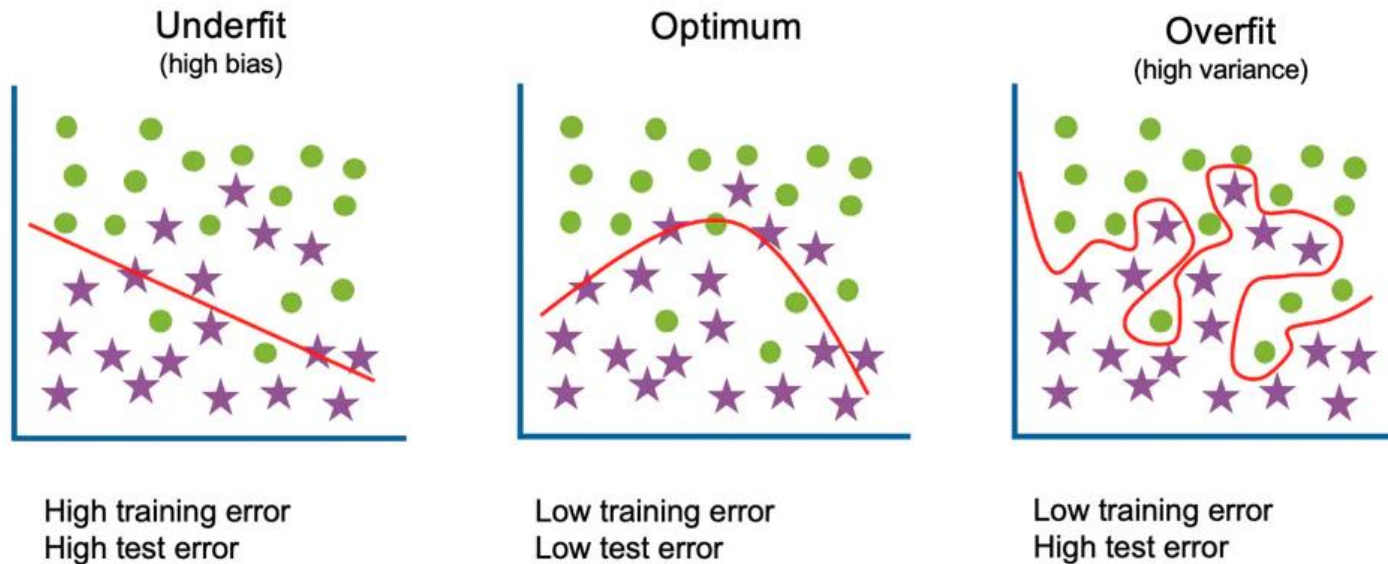


THE WAYS TO AVOID OVERFITTING

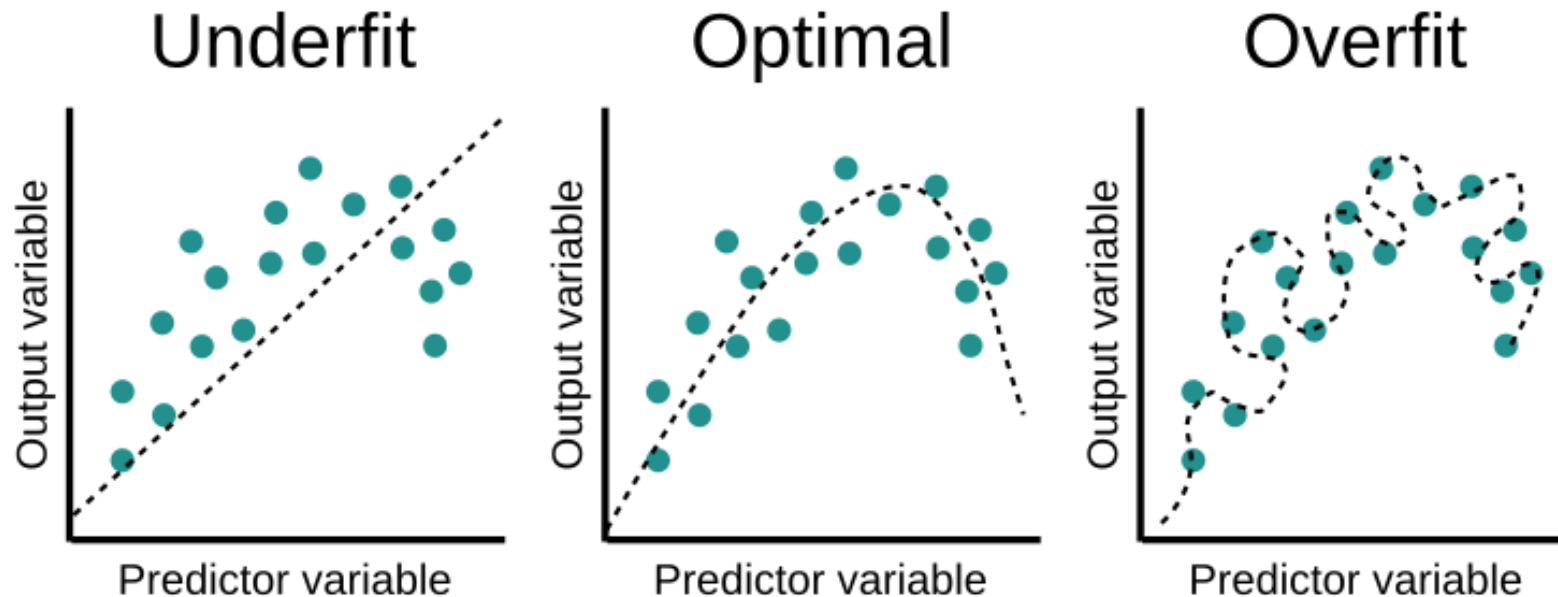
What is Overfitting?

- Overfitting occurs when a model fits **exactly** against its training data.



Why we don't want this?

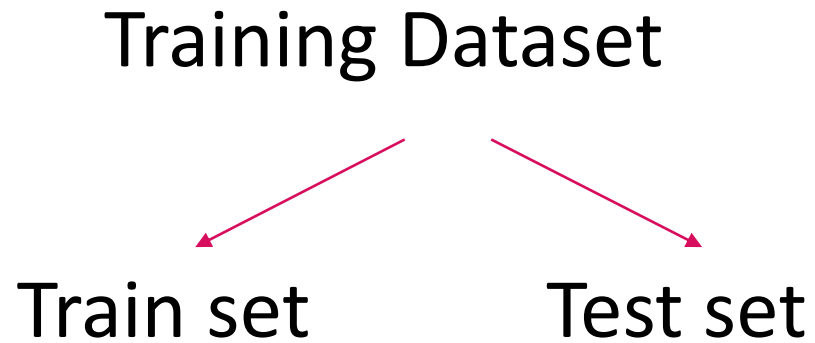
- Because the model cannot perform accurately against unseen data. **Generalization of a model** is what we need to make accurate predictions.



Why does this happen?

- When the model trains on data for **too long**
- When the model is **too complex**
- When the model **memorize** the 'noise'

How can we understand Overfitting?



Training set has **low error rate**
Test set has **high error rate**

} Overfitting

How to avoid Overfitting?

- Feature selection
- Train with more data
- Data augmentation
- Regularization

Feature Selection

- Sometimes some features are more redundant comparing the others. We have to identify the **most relevant features** and train the model with them.

Train with more data

- Expanding the data set provide the model to solve the relation between the input and output.
- **Risk:** If the data we add is **noisy, irrelevant and not clean**, this yields more problem than a solution.

Data augmentation

- Sometimes, noisy data is added to make a model more stable.
- **Risk:** The **amount of the noise** we add is important in this method.

Regularization

- Ridge Regression
- Lasso Regression

Questions?